Chapter 9
Database Management Systems



Accounting Information Systems, 7e
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Objectives for Chapter 9

- Understand the operational problems inherent in the flat-file approach to data management that gave rise to the database concept.
- Understand the relationships among the defining elements of the database environment.
- Understand the anomalies caused by unnormalized databases and the need for data normalization.
- Be familiar with the stages in database design, including entity identification, data modeling, constructing the physical database, and preparing user views.
- Be familiar with the operational features of distributed databases and recognize the issues that need to be considered in deciding on a particular database configuration.

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Flat-File Versus Database Environments

- Computer processing involves two components: data and instructions (programs).
- Conceptually, there are two methods for designing the interface between program instructions and data:
  - *File-oriented processing*: A specific data file was created for each application.
  - *Data-oriented processing*: Create a single data repository to support numerous applications.
- Disadvantages of file-oriented processing include
  - redundant data and programs
  - varying formats for storing the redundant data

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Flat-File Data Management

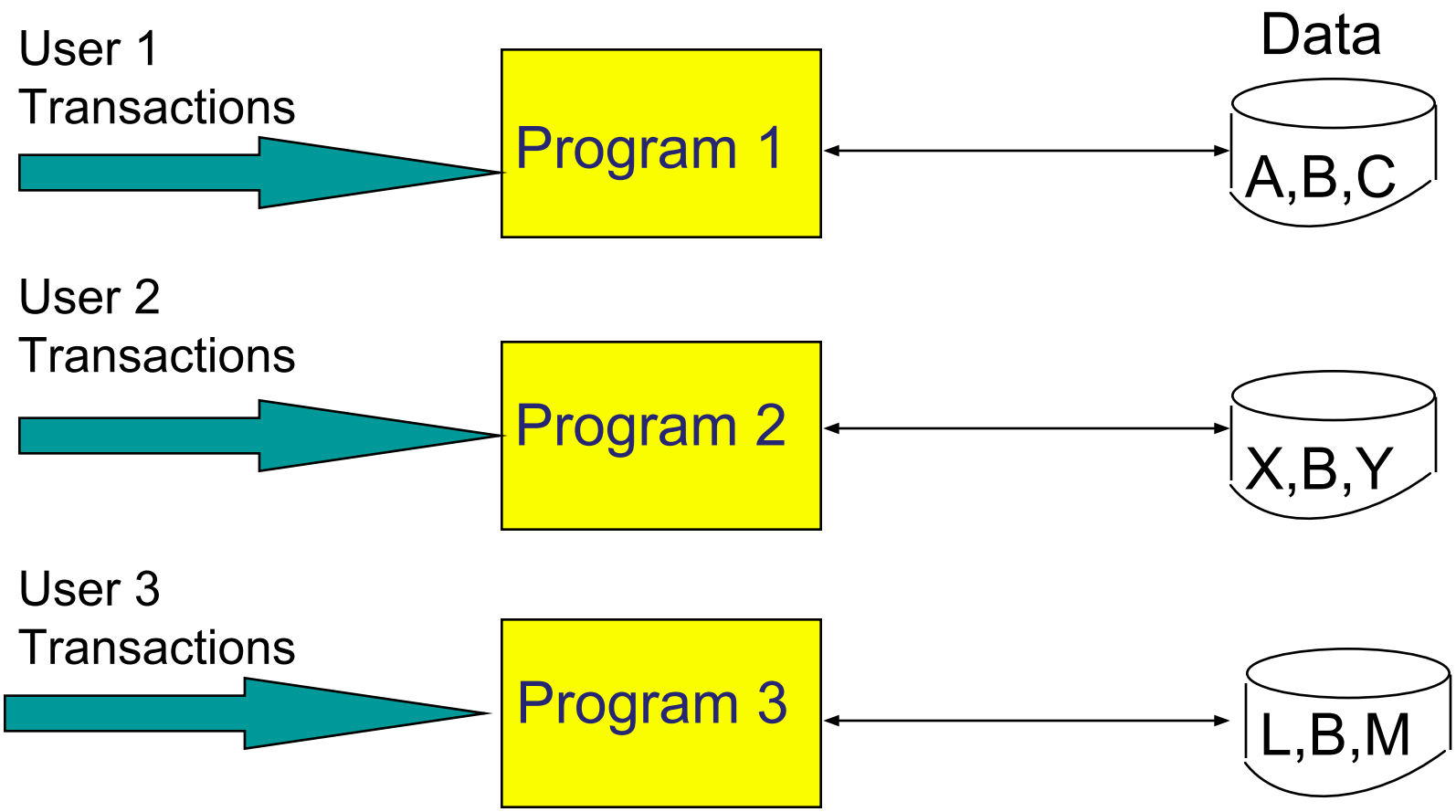


Figure 9-1

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Data Redundancy and Flat-File Problems

- **Data Storage -** creates excessive storage costs of paper documents and/or magnetic form.
- **Data Updating -** any changes or additions must be performed multiple times.
- **Currency of Information** – has the potential problem of failing to update all affected files.
- **Task-Data Dependency -** user unable to obtain additional information as his or her needs change

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The Database Concept

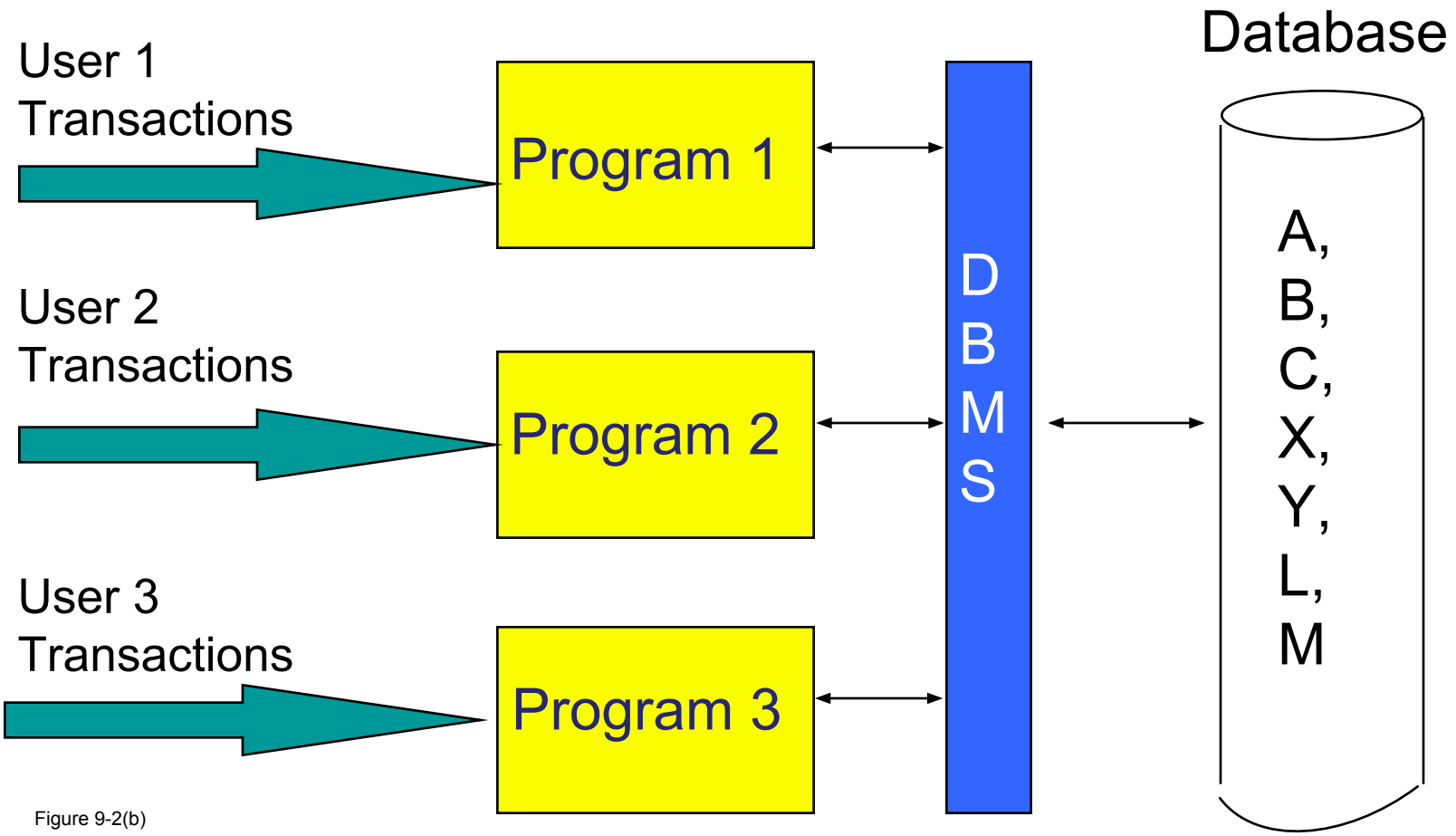


Figure 9-2(b)

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Advantages of the Database Approach

Data sharing/centralized database resolves flat-file problems:

- *No data redundancy:* Data is stored only once, eliminating data redundancy and reducing storage costs.
- *Single update:* Because data is in only one place, it requires only a single update, reducing the time and cost of keeping the database current.
- *Current values:* A change to the database made by any user yields current data values for all other users.
- *Task-data independence:* As users' information needs expand, the new needs can be more easily satisfied than under the flat-file approach.

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Disadvantages of the Database Approach

- Can be costly to implement
  - additional hardware, software, storage, and network resources are required.
- Can only run in certain operating environments
  - may make it unsuitable for some system configurations.
- Because it is so different from the file-oriented approach, the database approach requires training users
  - may be inertia or resistance.

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Elements of the Database Environment

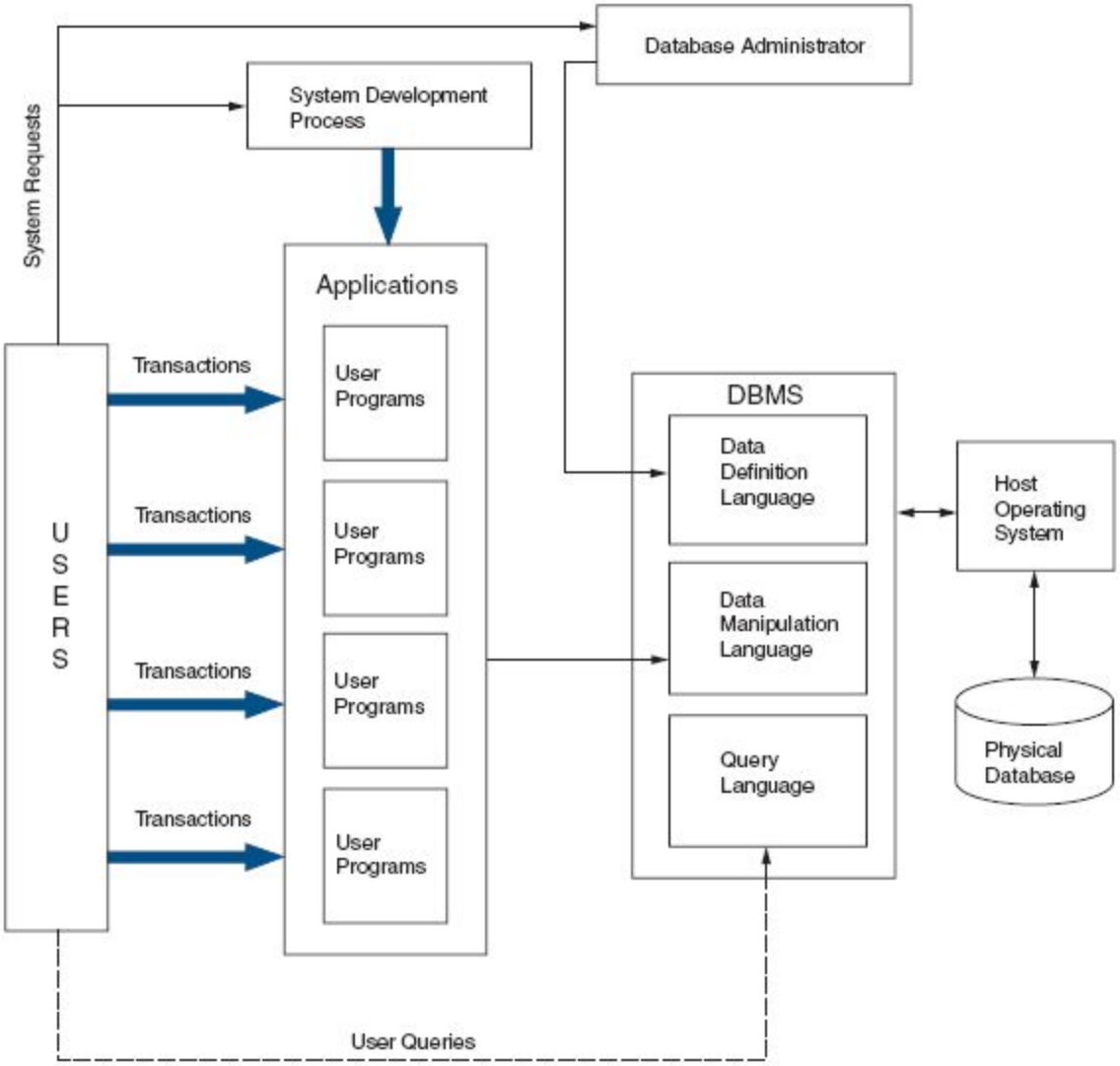


Figure 9-3

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Internal Controls and DBMS

- The database management system stands between the user and the database *per se.*
- Thus, commercial DBMS's (e.g., Access or Oracle) actually consist of a database *plus…*
  - software to manage the database, especially controlling access and other internal controls
  - software to generate reports, create data-entry forms, etc.
- The DBMS has special software to control which data elements each user is authorized to access.

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DBMS Features

- **Program Development -** user created applications
- **Backup and Recovery** - copies database.
- **Database Usage Reporting** - captures statistics on database usage (who, when, etc.).
- **Database Access** - authorizes access to sections of the database.
- Also…
  - **User Programs** - makes the presence of the DBMS transparent to the user.
  - **Direct Query** - allows authorized users to access data without programming.

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Data Definition Language (DDL)

- DDL is a programming language used to define the database *per se*.
  - It identifies the names and the relationship of all data elements, records, and files that constitute the database.
- DDL defines the database on three viewing levels
  - **Internal view** – physical arrangement of records (*1 view*)
  - **Conceptual view (schema)** – representation of database (*1 view*)
  - **User view (subschema)** – the portion of the database each user views (*many views*)

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Overview of DBMS Operation

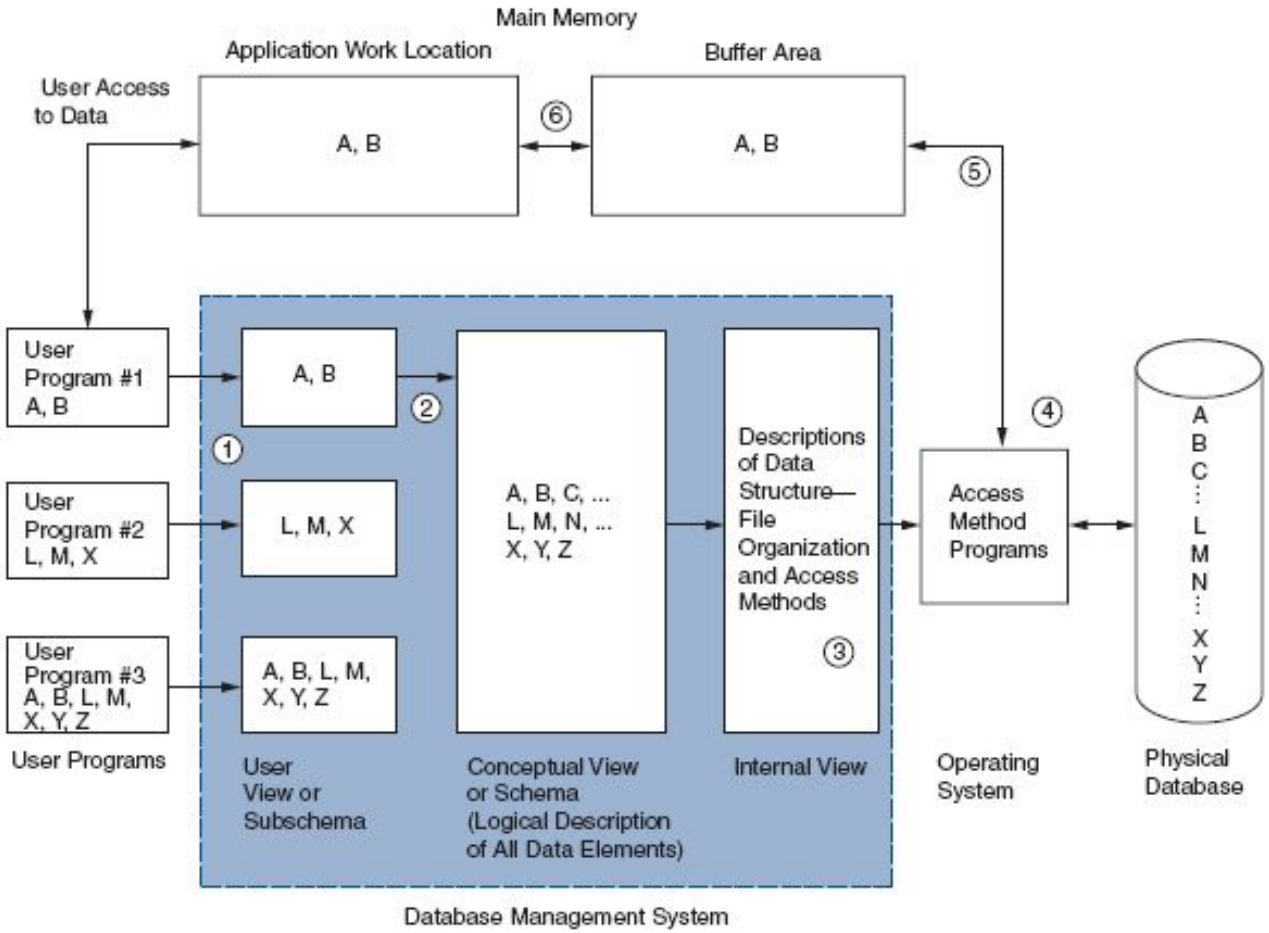


Figure 9-4

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Data Manipulation Language (DML)

- DML is the proprietary programming language that a particular DBMS uses to retrieve, process, and store data to / from the database.
- Entire user programs may be written in the DML, or selected DML commands can be inserted into universal programs, such as COBOL and FORTRAN.
- Can be used to 'patch' third party applications to the DBMS

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Query Language

- The query capability permits end users and professional programmers to access data in the database without the need for conventional programs.
  - Can be an internal control issue since users may be making an 'end run' around the controls built into the conventional programs
- IBM's *structured query language (SQL)* is a fourth-generation language that has emerged as the standard query language.
  - Adopted by ANSI as the standard language for *all* relational databases

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Functions of the DBA

Database Planning:

Develop organization's database strategy
Define database environment
Define data requirements
Develop data dictionary

Design:

Logical database (schema)
External users' views (subschemas)
Internal view of database
Database controls

Implementation:

Determine access policy
Implement security controls
Specify test procedures
Establish programming standards

Operation and Maintenance:

Evaluate database performance
Reorganize database as user needs demand
Review standards and procedures

Change and Growth:

Plan for change and growth
Evaluate new technology

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Database Conceptual Models

- Refers to the particular method used to organize records in a database.
  - a.k.a. “logical data structures”
- *Objective:* develop the database efficiently so that data can be accessed quickly and easily.
- There are three main models:
  - hierarchical (tree structure)
  - network
  - relational
- Most existing databases are relational. Some legacy systems use hierarchical or network databases.

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The Relational Model

- The relational model portrays data in the form of two dimensional ‘tables’.
- Its strength is the ease with which tables may be linked to one another.
  - a major weakness of hierarchical and network databases
- Relational model is based on the relational algebra functions of restrict, project, and join.

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The Relational Algebra Functions Restrict, Project, and Join

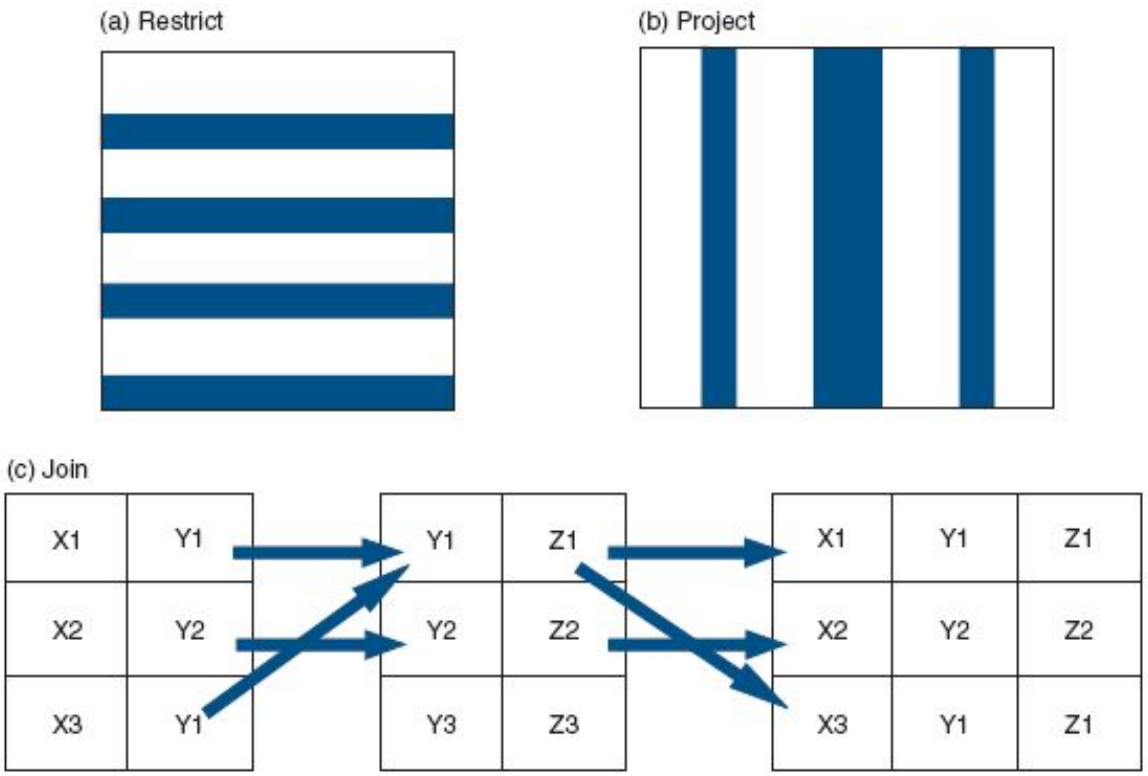


Figure 9-9

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Associations and Cardinality

- **Association**
  - Represented by a line connecting two entities
  - Described by a verb, such as ships, requests, or receives
- **Cardinality** – the degree of association between two entities
  - The number of possible occurrences in one table that are associated with a single occurrence in a related table
  - Used to determine *primary keys* and *foreign keys*

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Examples of Entity Associations

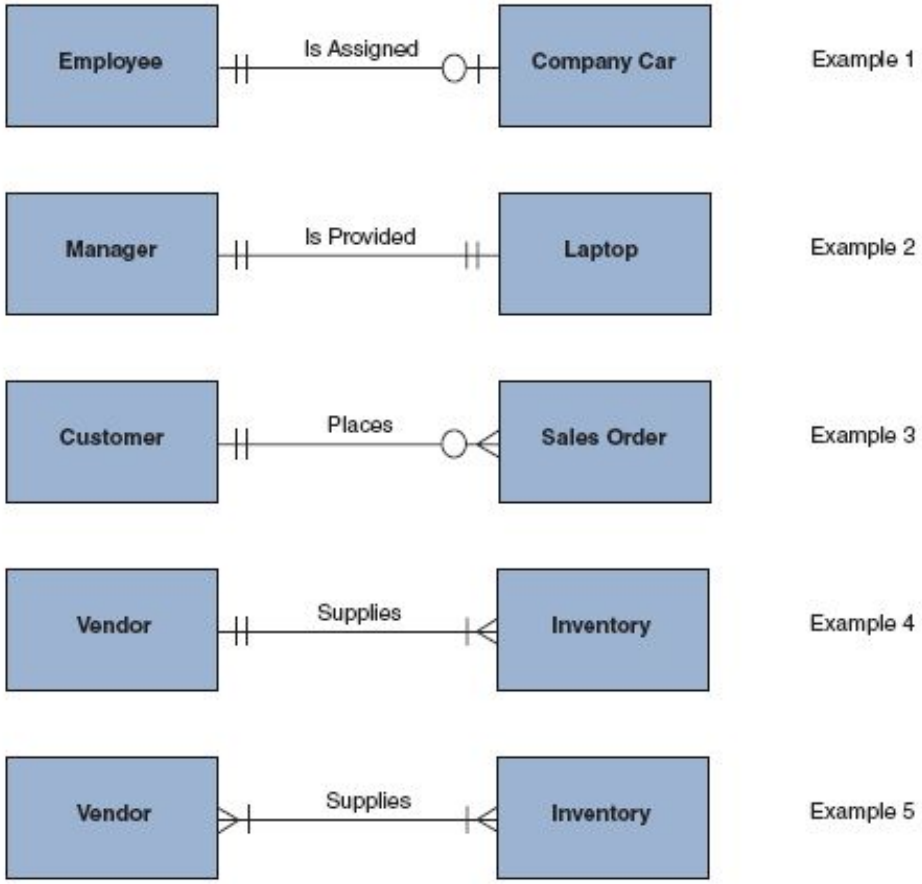


Figure 9-11

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Properly Designed Relational Tables

- Each row in the table must be unique in at least one attribute, which is the primary key.
  - Tables are linked by embedding the primary key into the related table as a foreign key.
- The attribute values in any column must all be of the same class or data type.
- Each column in a given table must be uniquely named.
- Tables must conform to the rules of normalization, i.e., free from structural dependencies or anomalies.

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Three Types of Anomalies

- **Insertion Anomaly**: A new item cannot be added to the table until at least one entity uses a particular attribute item.
- **Deletion Anomaly**: If an attribute item used by only one entity is deleted, all information about that attribute item is lost.
- **Update Anomaly**: A modification on an attribute must be made in each of the rows in which the attribute appears.
- Anomalies can be corrected by creating additional **relational tables**.

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Advantages of Relational Tables

- Removes all three types of anomalies.
- Various items of interest (customers, inventory, sales) are stored in separate tables.
- Space is used efficiently.
- Very flexible – users can form ad hoc relationships.

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The Normalization Process

- A process which **systematically splits unnormalized complex tables into smaller tables** that meet two conditions:
  - all nonkey (secondary) attributes in the table are dependent on the primary key
  - all nonkey attributes are independent of the other nonkey attributes
- When unnormalized tables are split and reduced to third normal form, they must then be linked together by foreign keys.

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Steps in the Normalization Process

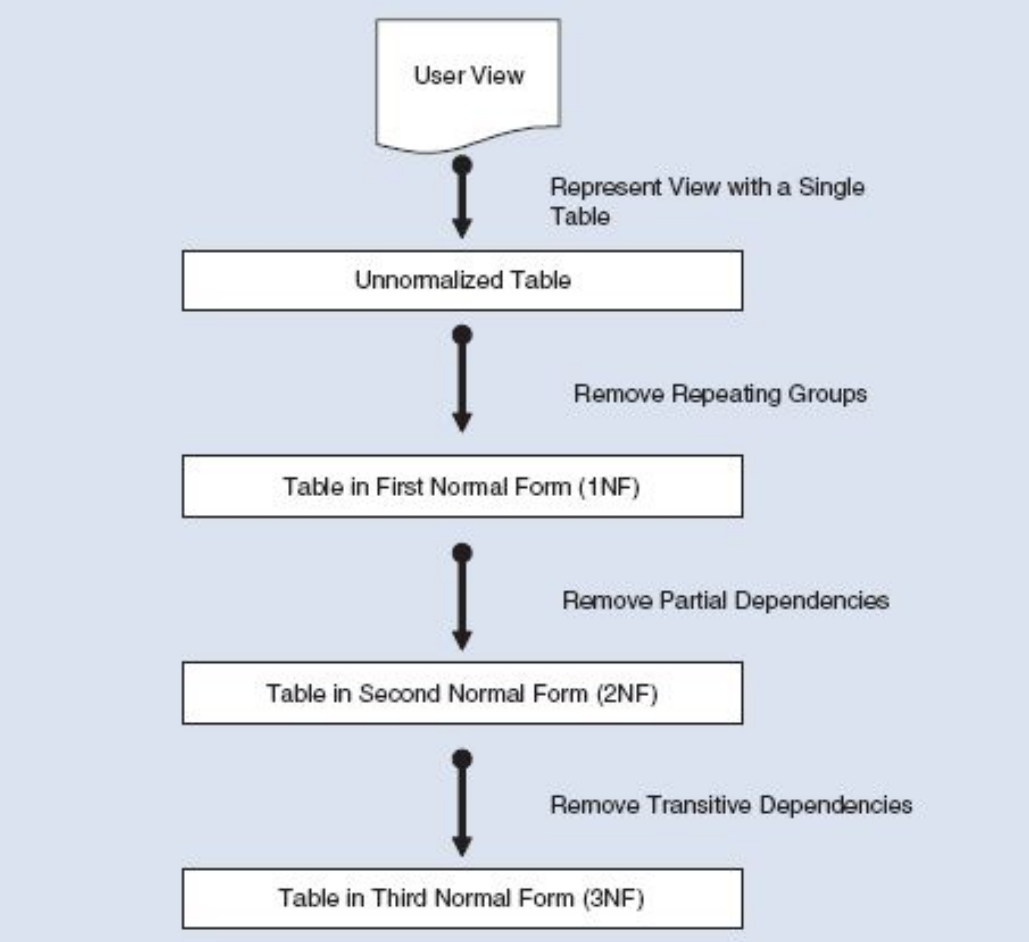


Figure 9-34

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Accountants and Data Normalization

- *Update anomalies* can generate **conflicting and obsolete database values**.
- *Insertion anomalies* can result in **unrecorded transactions and incomplete audit trails**.
- *Deletion anomalies* can cause the loss of accounting records and the **destruction of audit trails.**
- *Accountants should understand the data normalization process and be able to determine whether a database is properly normalized.*

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Six Phases in Designing Relational Databases

1. Identify entities
   - identify the primary entities of the organization
   - construct a data model of their relationships
2. Construct a data model showing entity associations
   - determine the associations between entities
   - model associations into an ER diagram

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Six Phases in Designing Relational Databases

3. Add primary keys and attributes
   - assign primary keys to all entities in the model to uniquely identify records
   - every attribute should appear in one or more user views
4. Normalize and add foreign keys
   - remove repeating groups, partial and transitive dependencies
   - assign foreign keys to be able to link tables

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Six Phases in Designing Relational Databases

5. Construct the physical database
   - create physical tables
   - populate tables with data
6. Prepare the user views
   - normalized tables should support all required views of system users
   - user views restrict users from having access to unauthorized data

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Distributed Data Processing (DDP)

- Data processing is organized around several information processing units (IPUs) distributed throughout the organization.
  - Each IPU is placed under the control of the end user.
- DDP does not always mean total decentralization.
  - IPUs in a DDP system are still connected to one another and coordinated.
  - Typically, DDP's use a centralized database.
  - Alternatively, the database can be distributed, similar to the distribution of the data processing capability.

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Centralized Databases in DDP Environment

- The data is retained in a central location.
- Remote IPUs send requests for data.
- Central site services the needs of the remote IPUs.
- *The actual processing of the data is performed at the remote IPU.*

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Advantages of DDP

- **Cost reductions** in hardware and data entry tasks
- **Improved cost control** responsibility
- **Improved user satisfaction** since control is closer to the user level
- **Backup of data** can be improved through the use of multiple data storage sites

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Disadvantages of DDP

- Loss of control
- Mismanagement of resources
- Hardware and software incompatibility
- Redundant tasks and data
- Consolidating incompatible tasks
- Difficulty attracting qualified personnel
- Lack of standards

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Data Currency

- Occurs in DDP with a centralized database
- During transaction processing, data will temporarily be inconsistent as records are read and updated.
- **Database lockout** procedures are necessary to keep IPUs from reading inconsistent data and from writing over a transaction being written by another IPU.

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Distributed Databases: Partitioning

- Splits the central database into segments that are distributed to their primary users.
- Advantages:
  - users' control is increased by having data stored at local sites.
  - transaction processing response time is improved.
  - volume of transmitted data between IPUs is reduced.
  - reduces the potential data loss from a disaster.

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The Deadlock Phenomenon

- Especially a problem with partitioned databases
- Occurs when multiple sites lock each other out of data that they are currently using.
  - One site needs data locked by another site.
- Special software is needed to analyze and resolve conflicts.
  - Transactions may be terminated and restarted.

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The Deadlock Condition

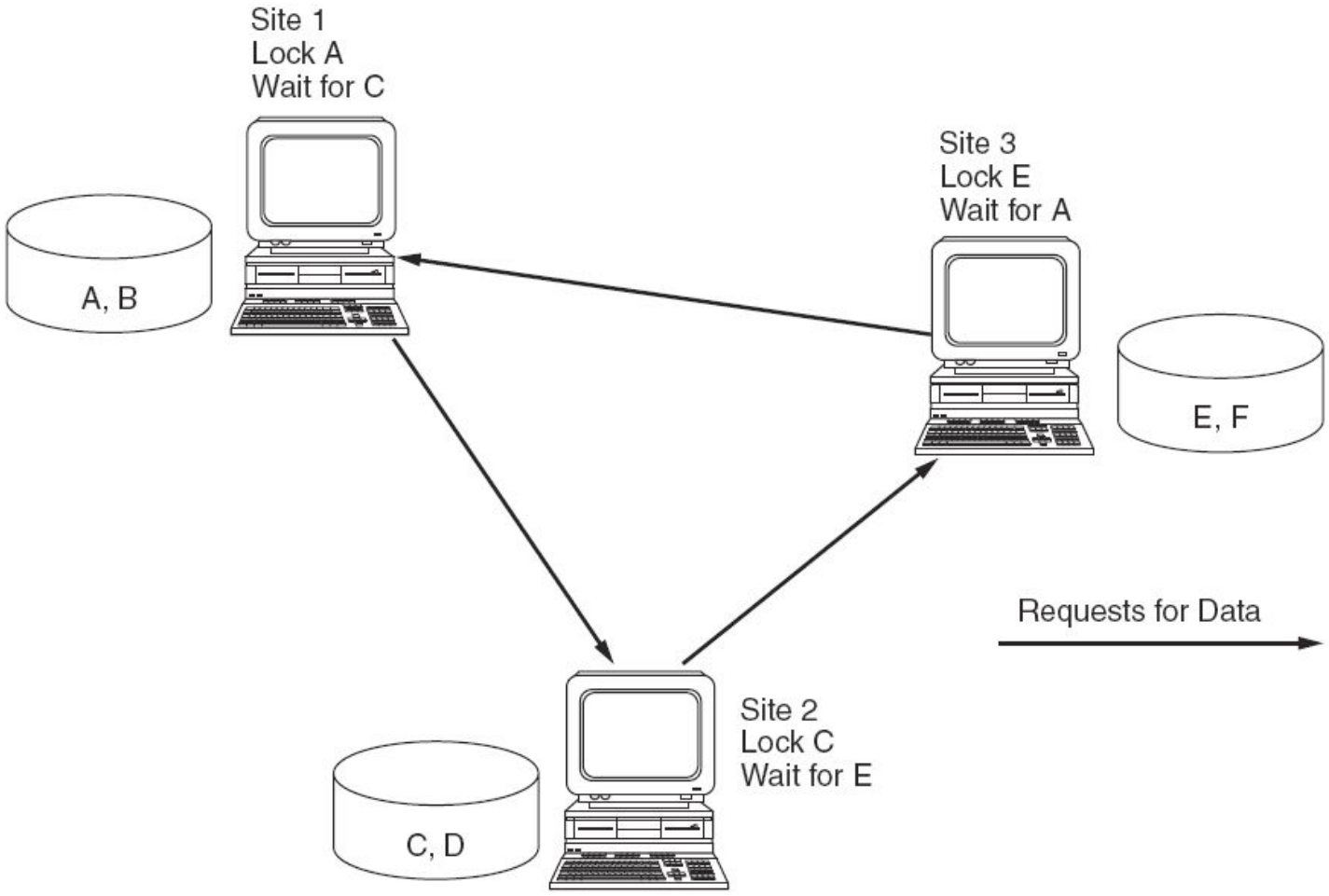


Figure 9-26

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Distributed Databases: Replication

- The duplication of the entire database for multiple IPUs
- Effective for situations with a high degree of data sharing, but no primary user
  - Supports read-only queries
- Data traffic between sites is reduced considerably.

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Concurrency Problems and Control Issues

- Database concurrency is the presence of complete and accurate data at all IPU sites.
- With replicated databases, maintaining current data at all locations is difficult.
- Time stamping is used to serialize transactions.
  - Prevents and resolves conflicts created by updating data at various IPUs.

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Distributed Databases and the Accountant

- The following database options impact the organization's ability to maintain database integrity, to preserve audit trails, and to have accurate accounting records.
  - Centralized or distributed data?
  - If distributed, replicated or partitioned?
  - If replicated, total or partial replication?
  - If partitioned, what is the allocation of the data segments among the sites?

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